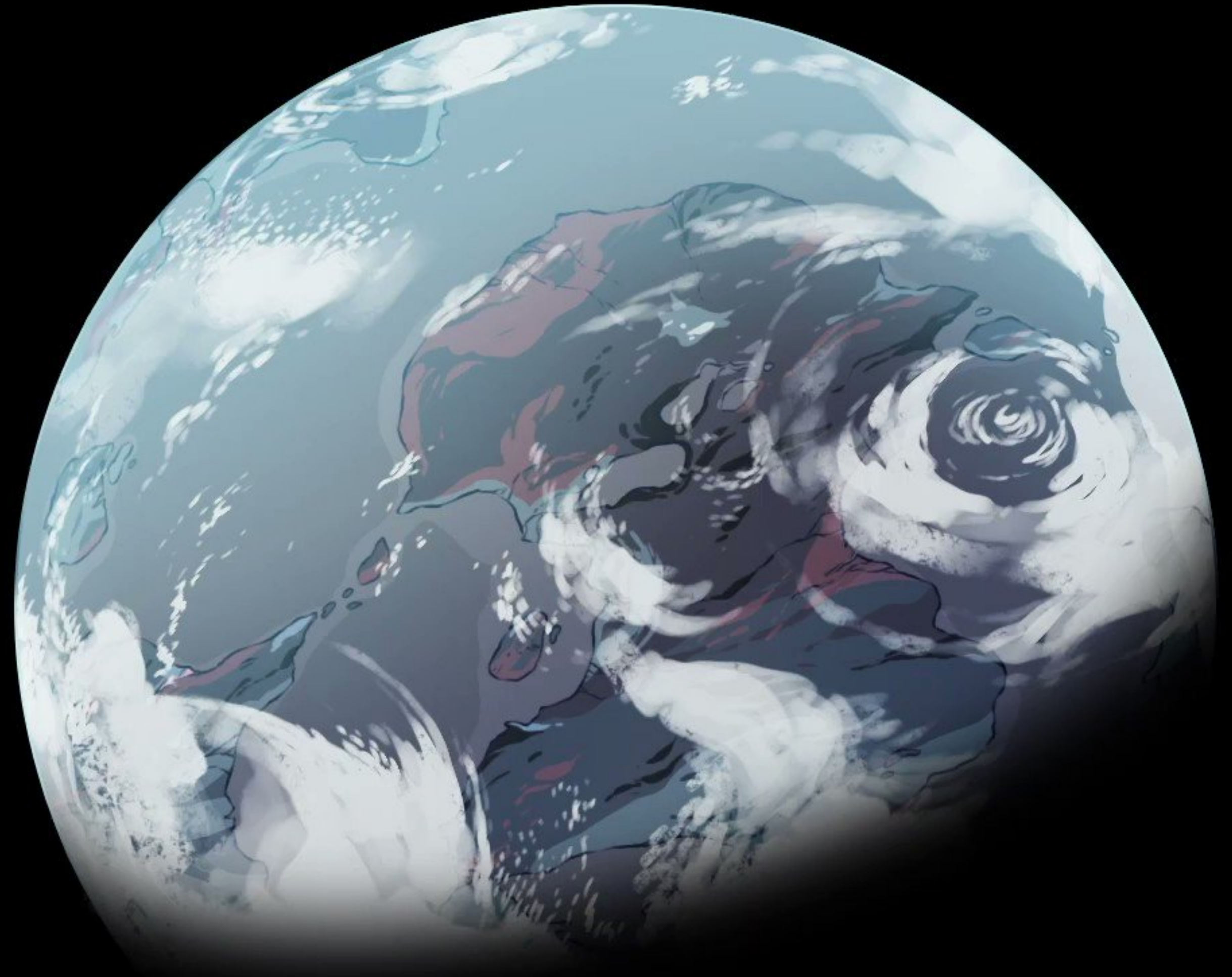


Science & Science Fiction

33-120 | Prof. Katie Breivik



- ▶ Quiz 5 is today!! — SETI
 - ▶ 30 pts available; Due @ midnight
- ▶ SETI Team Project Due Wednesday, April 2!

- ▶ Today Begins: What does it mean to be Human?
 - ▶ Insights from Mary Shelly's *Frankenstein*
 - ▶ Transplants, Spread of infectious disease
 - ▶ Cell structure, DNA, cloning, body modification
 - ▶ Human teleportation

Extra Credit Opportunity 5!



Course Feedback Survey due
Wednesday, April 2 at Midnight!
25 points available!!

<https://tinyurl.com/SciSciFi-EC5>

Frankenstein

Mary Wollstonecraft Shelley

First edition: 1818

Second edition: 1823

Considered by many to be *the* first
work of science fiction

Tells the story of a young scientist, Victor
Frankenstein, who becomes jaded by
university studies and forges his own path

FRANKENSTEIN ;

OR,

THE MODERN PROMETHEUS.

IN THREE VOLUMES.

Did I request thee, Maker, from my clay
To mould me man ? Did I solicit thee
From darkness to promote me ?—

PARADISE LOST.

VOL. I.

LONDON :

PRINTED FOR

LACKINGTON, HUGHES, HARDING, MAJOR, & JONES,
FINSBURY SQUARE.

1818.

Concepts to consider from Frankenstein

1. Innate human curiosity about the natural and/or physical world
2. Scientific hubris and the 'God complex' of a mad scientist
3. Dangerous medicine
4. Human longing for relationships

FRANKENSTEIN ;

OR,

THE MODERN PROMETHEUS.

IN THREE VOLUMES,

Did I request thee, Maker, from my clay
To mould me man? Did I solicit thee
From darkness to promote me?—

PARADISE LOST.

VOL. I.

LONDON :
PRINTED FOR
LACKINGTON, HUGHES, HARDING, MAJOR, & JONES,
FINSBURY SQUARE.

1818.

Innate human curiosity about the natural and/or physical world



A young Frankenstein witnesses an oak tree being blasted to splinters by lightning

“Could this tremendous power be harnessed to create life?”

This experience inspires him to go to medical school and study the scientific processes of life

A professor further mentors/inspires his curiosity

Innate human curiosity about the natural and/or physical world



A young Frankenstein witnesses an oak tree being blasted to splinters by lightning

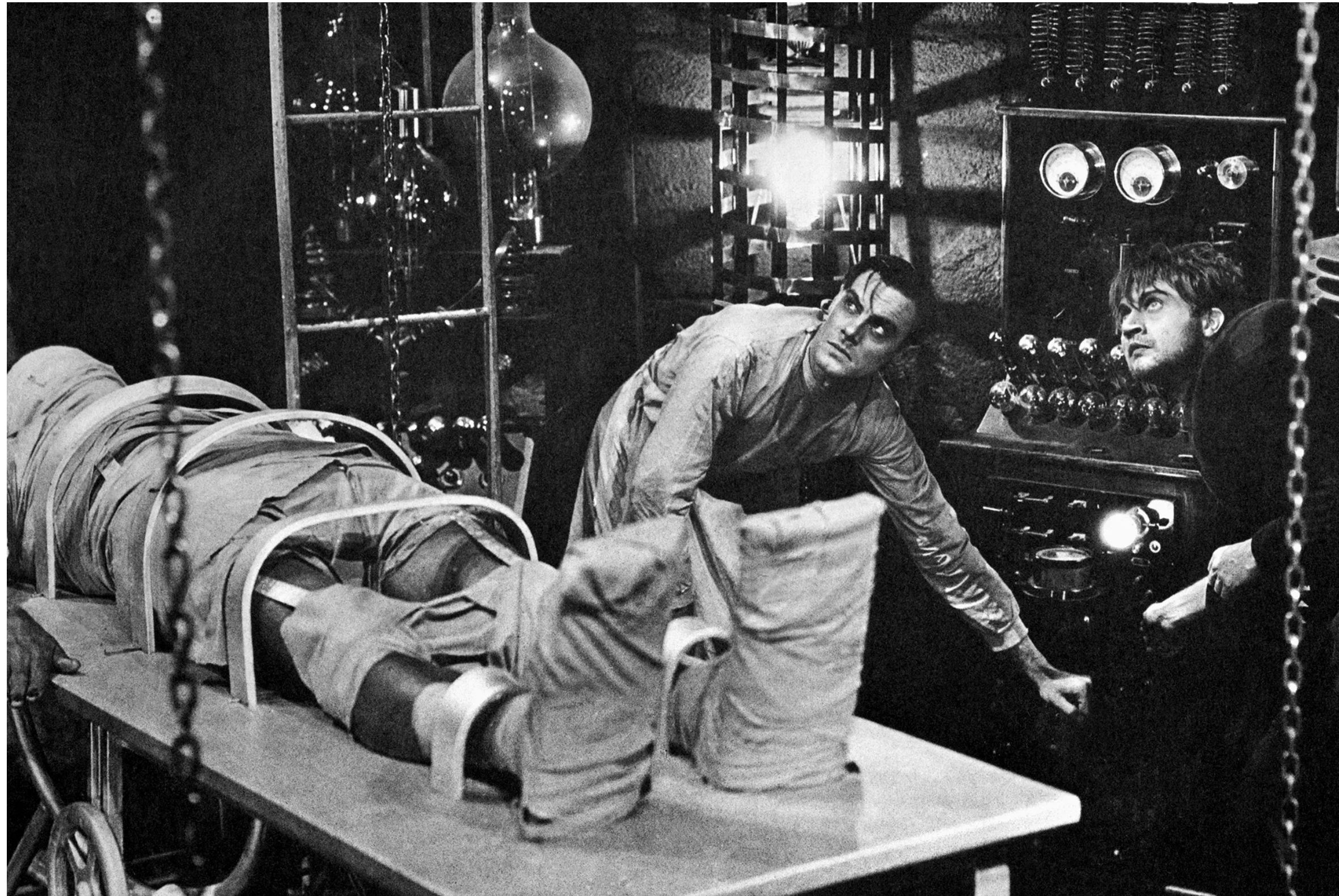
“Could this tremendous power be harnessed to create life?”

This experience inspires him to go to medical school and study the scientific processes of life

A professor further mentors/catalyzes his curiosity

<https://tinyurl.com/SciSciFi-28>

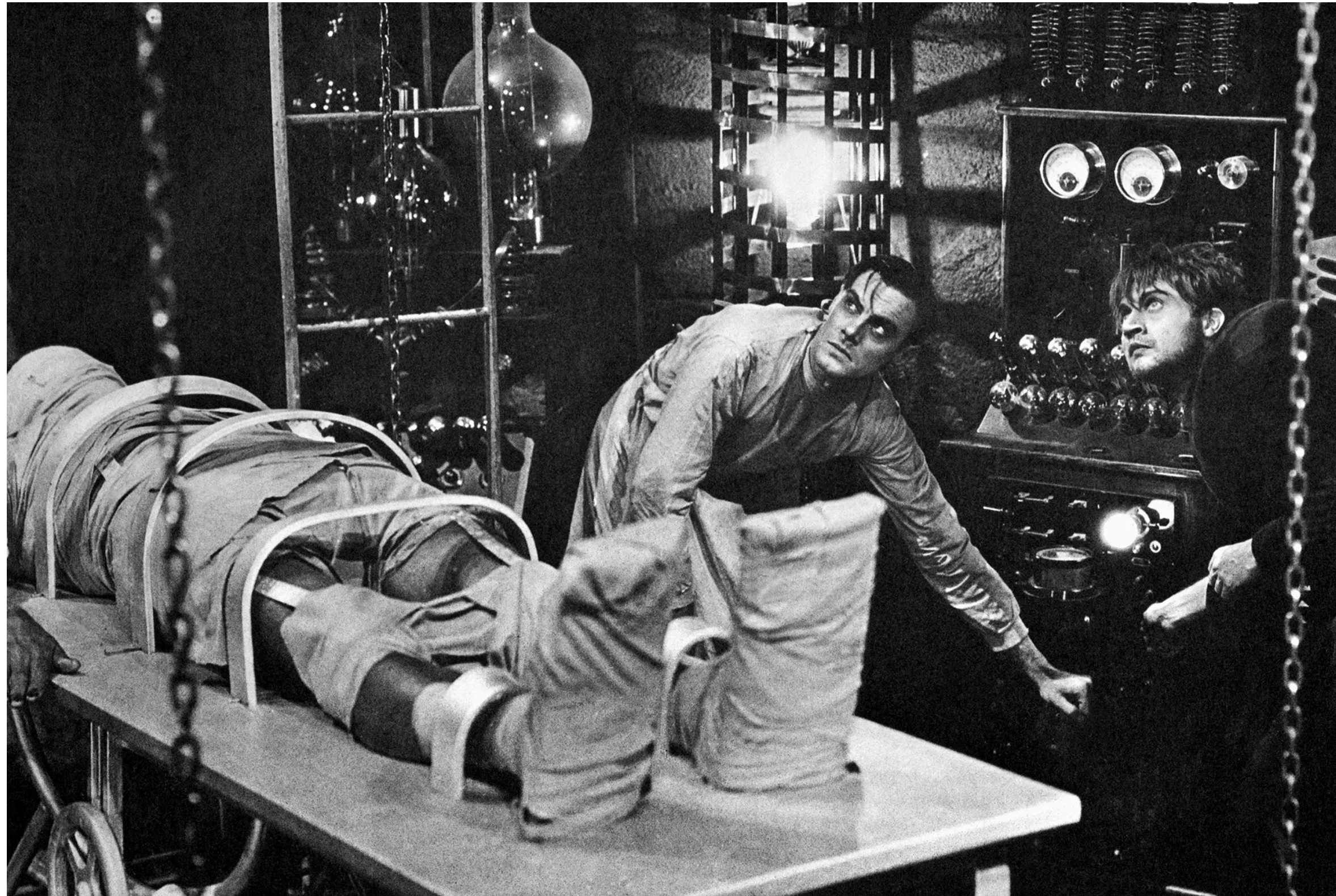
Scientific hubris and the 'God complex' of a mad scientist*



While in University, Frankenstein develops a method to infuse life into lifeless material

It was on a dreary night of November that I beheld the accomplishment of my toils. With an anxiety that almost amounted to agony, I collected the instruments of life around me that I might infuse a spark of being into the lifeless thing that lay at my feet. It was already one in the morning; the rain pattered dismally against the panes, and my candle was nearly burnt out, when, by the glimmer of the half-extinguished light, I saw the dull yellow eye of the creature open; it breathed hard, and a convulsive motion agitated its limbs.

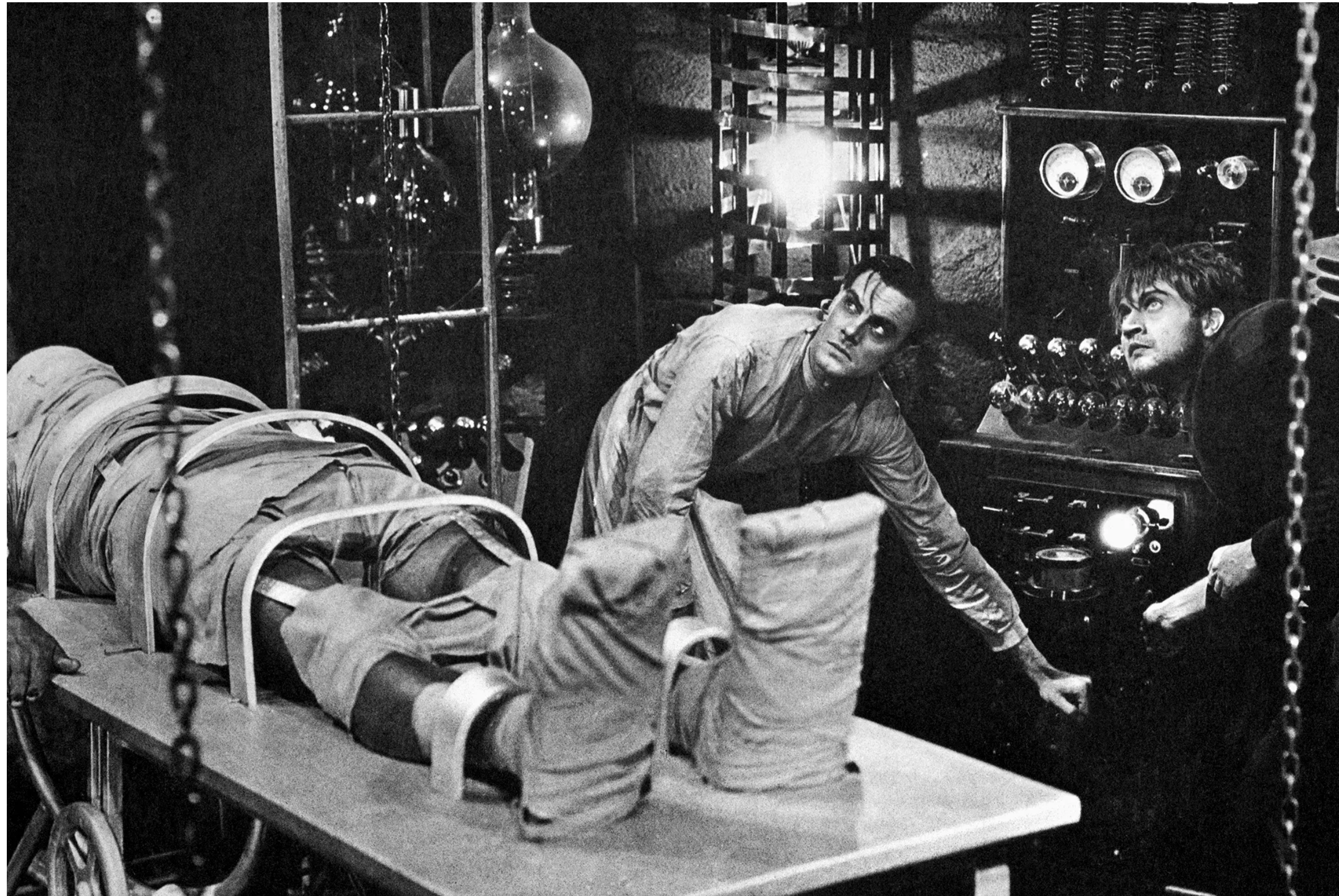
Scientific hubris and the 'God complex' of a mad scientist*



While in University, Frankenstein develops a method to infuse life into lifeless material

In many film adaptations, Frankenstein explicitly aims to create a 'beautiful' creature by selecting corpses with nice physical features — this backfires when the Creature is seen to be anything but beautiful by society

Scientific hubris and the 'God complex' of a mad scientist*

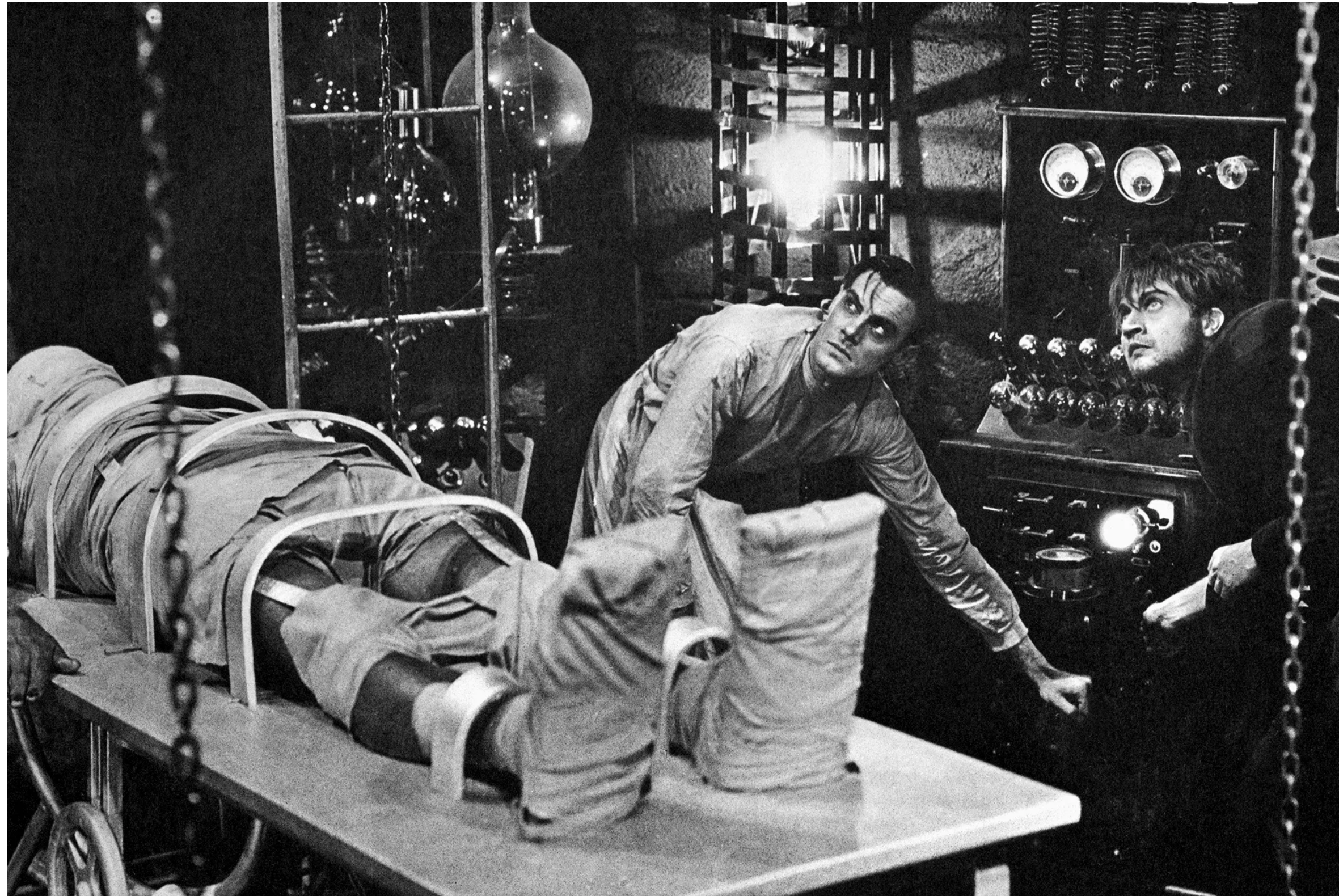


Frankenstein

directed: James Whale

Universal Studios (1931)

Scientific hubris and the ‘God complex’ of a mad scientist*



Frankenstein

The film was a victim of the
“Hays Code” written in 1930 and
enforced from 1936-1968

Edited out the lines:

“In the name of God! Now I know what
it feels like to be God!”

Dangerous Medicine

Invention of **anesthetic surgery** nor **antiseptic surgery** had not yet happened when Frankenstein was first published in 1818

anesthetic surgery: pioneered by William T.G. Morton in 1846

—> pain-free surgery: not necessarily more safe!

antiseptic surgery: pioneered by Joseph Lister in 1865 using carbolic acid

—> infection-free surgery: *definitely* more safe!

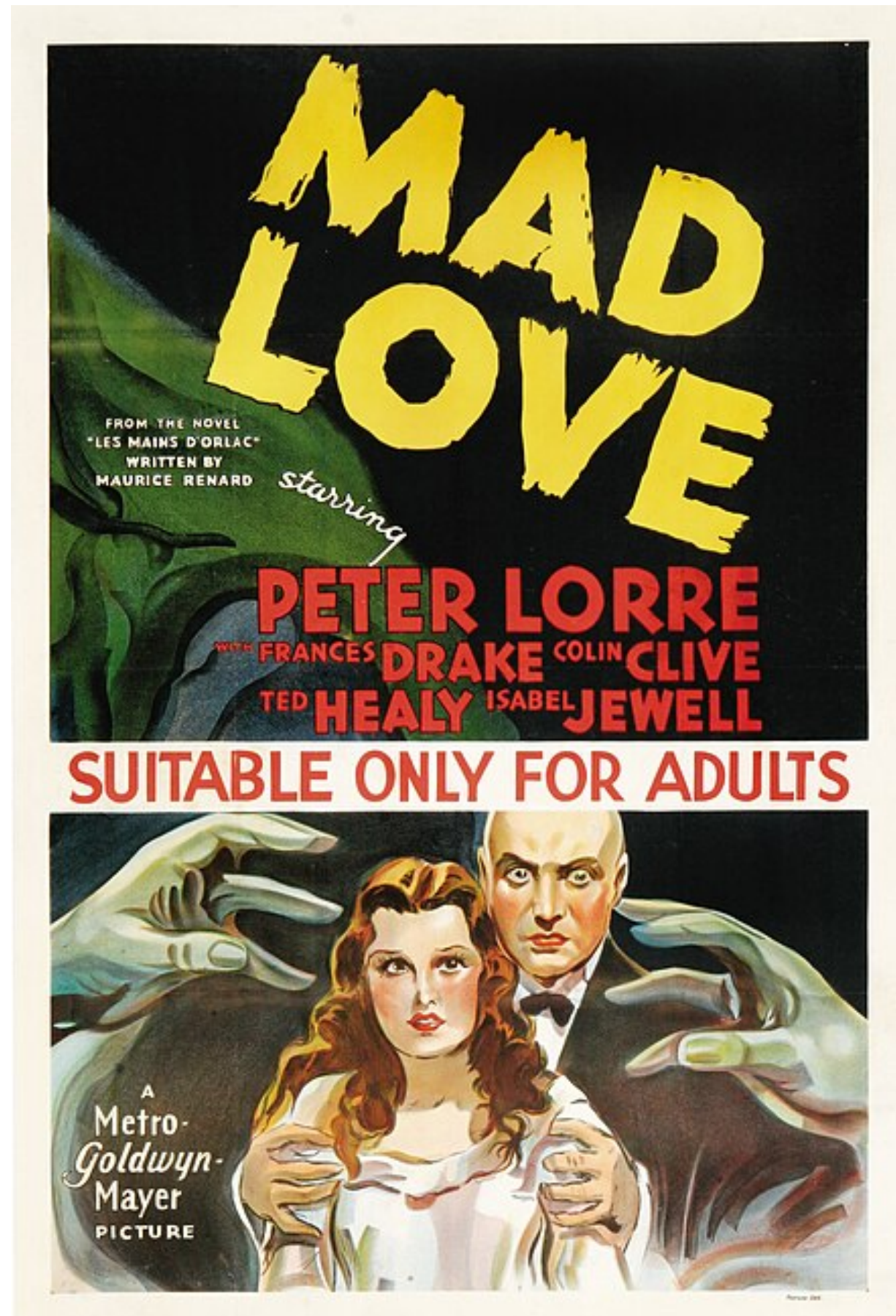
Human longing for relationships



A central tenet of both the book and all film depictions is the Creature's longing for emotional and physical relationships — and the consequences for being restricted from having them

Perhaps *Frankenstein* tells us more about sociology than science

Transplants in Science Fiction (Horror)



Mad Love

based on the novel

The Hands of Orlac

by Maurice Renard (1921)

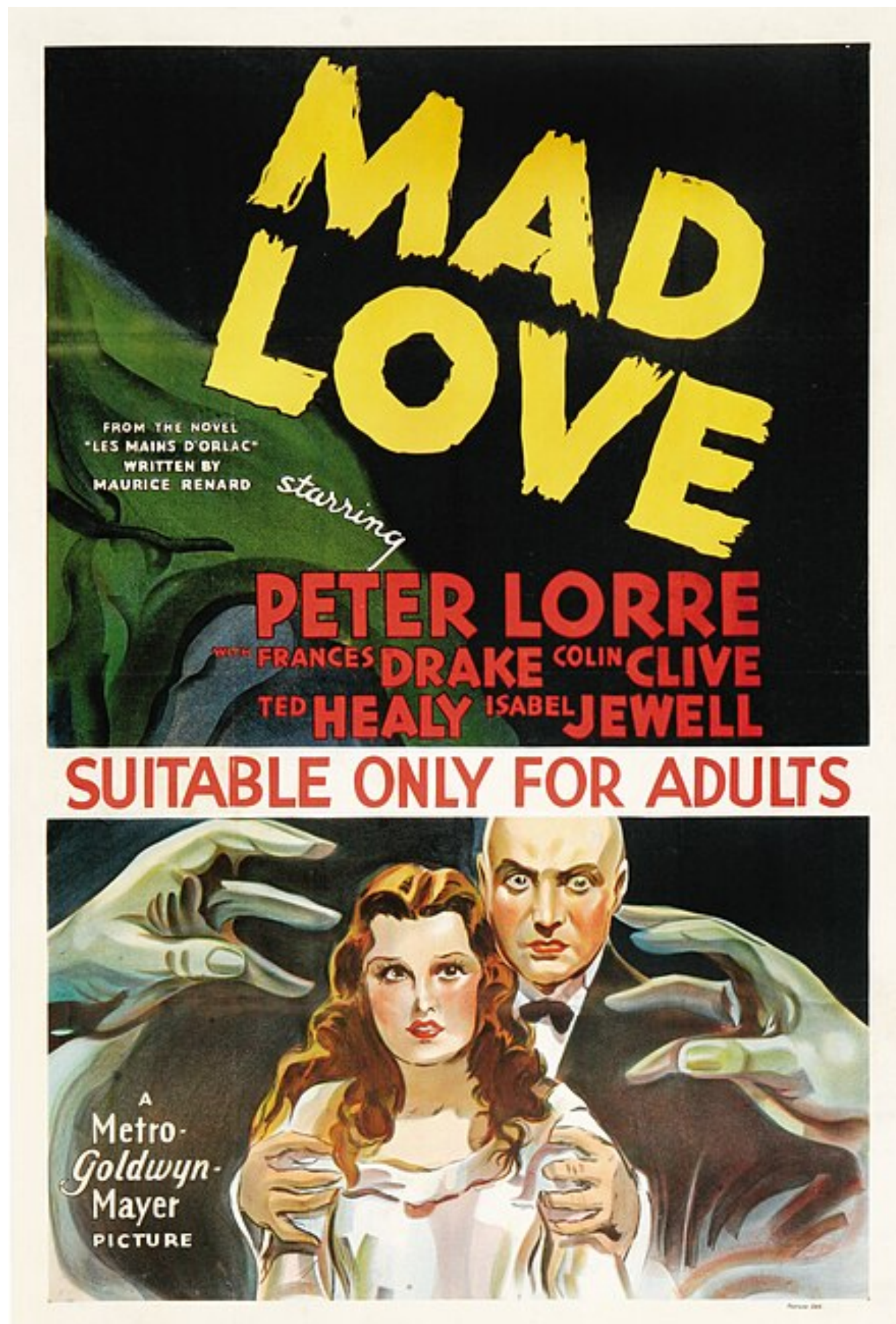
and Austrian film

The Hands of Orlac

directed by Robert Wiene (1924)

directed: Karl Freund

MGM (1935)



Mad Love's successful hand transplant predates even the earliest attempts for hand transplants in reality:

First attempted in
Clinica Guayaquil, Ecuador in 1964
(unsuccessful due to tissue rejection)

First successful hand transplant in
Lyon, France happened in 1998

The Man They Could Not Hang

directed: Nick Grinde

Columbia Pictures (1939)





First depiction of a
cardiopulmonary bypass
machine

Not used until 1951!

Hays code strikes again: no blood could be depicted

- ▶ Quiz 5 is today!! — SETI
 - ▶ 30 pts available
 - ▶ Due @ midnight
- ▶ Next time:
 - ▶ Finish dangerous medicine
 - ▶ Spread of infectious disease
 - ▶ Cell structure & Radiation